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## AWS Directory Service — Overview

### What is AWS Directory Service?

**AWS Directory Service** is a managed service that lets you set up and run **Microsoft Active Directory (AD)** in the AWS Cloud or connect your AWS resources to an existing on-premises AD.

 It enables authentication, authorization, and directory-based management of users, computers, and services — just like on-prem AD, but in the cloud.

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### Types of Directory Services

| Service                                                                                                            | Description                                                                                     |
|--------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
|  <b>AWS Managed Microsoft AD</b> | Fully managed, cloud-native Microsoft Active Directory. Built on real Windows Server AD.        |
|  <b>AD Connector</b>            | Proxy to redirect AWS requests to your on-premises AD.                                          |
|  <b>Simple AD</b>               | Lightweight, low-cost directory powered by Samba. Good for dev/test. (Deprecated for new users) |

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### 1. AWS Managed Microsoft AD

 **Best for:** Enterprises wanting a fully functional AD in the cloud.

#### Features:

- Join EC2 instances to the domain.
- Enable **Group Policy**, **LDAP**, **Kerberos**, and **DNS**.
- Use for **Amazon RDS SQL Server** authentication.

- Integrates with **AWS IAM Identity Center**, WorkSpaces, and more.

#### **Deployment options:**

- **Standard Edition:** ~5000 users, 1GB storage
  - **Enterprise Edition:** ~100,000 users, 17GB storage
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## **2. AD Connector**

 **Best for:** Organizations with **existing on-prem AD**.

#### **How it works:**

- AD Connector acts as a **proxy**.
- No data is stored in AWS.
- AWS services authenticate via AD Connector → On-prem AD.

 Connects AWS services like:

- Amazon WorkSpaces
  - Amazon QuickSight
  - AWS IAM Identity Center
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## **3. Simple AD**

 **Best for:** Small, low-budget, non-enterprise dev/test use cases.

- Built on **Samba 4** (open-source AD-compatible).
- Supports basic AD features: user accounts, groups, and policies.

#### **Limitations:**

- No support for trust relationships
  - No schema extensions
  - Not recommended for production
  - Deprecated for new users
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#### **Use Cases**

| Use Case                                                                                                | Description                                  |
|---------------------------------------------------------------------------------------------------------|----------------------------------------------|
|  User Authentication | Central login for WorkSpaces, EC2, RDS, apps |
|  On-Prem Integration | Extend AD to AWS without duplication         |
|  File Services       | Use with Amazon FSx for Windows File Server  |
|  Identity Federation | Use AD groups for controlling access         |
|  Domain Join         | Auto-join EC2 Windows/Linux instances to AD  |

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#### **Lab Scenario: Deploy AWS Managed Microsoft AD**

 **Objective:** Create a managed AD and join an EC2 instance

**Step 1: Create VPC with 2 private subnets (different AZs)**

**Step 2: Launch AWS Directory Service**

1. Go to **Directory Service > Set up Directory**

2. Choose **AWS Managed Microsoft AD**

3. Provide:

- Directory name: `corp.example.com`
- Admin password
- VPC & subnets

 Wait for provisioning (can take ~20 mins)

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### Step 3: Launch a Windows EC2 instance in the same VPC

- Choose domain-join option during launch
  - Use the domain DNS name (`corp.example.com`)
  - Log in using AD credentials after joining
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### Step 4: Optional - Add a group policy or user via AD Tools

1. RDP into the EC2 instance
  2. Open **Active Directory Users and Computers**
  3. Create users, groups, or apply policies
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- Supports **multi-factor authentication (MFA)**
- Encrypted communication between domain controllers
- Integrated with **CloudTrail** for audit logging
- **Highly available** with multi-AZ deployments

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### Integration Highlights

| AWS Service                                                                                          | Benefit                             |
|------------------------------------------------------------------------------------------------------|-------------------------------------|
|  EC2                | Domain join, LDAP login             |
|  FSx for Windows    | Share access using AD credentials   |
|  WorkSpaces        | Central auth using corporate AD     |
|  QuickSight       | Authenticate users via AD Connector |
|  RDS (SQL Server) | Use Windows Authentication          |

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### Summary

| Feature                                                                                          | Details                                             |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|  Purpose      | Directory-based authentication and management       |
|  Main Types   | AWS Managed AD, AD Connector, Simple AD             |
|  Security     | MFA, auditing, encryption                           |
|  Connectivity | VPC integration, on-prem bridge (AD Connector)      |
|  Use Cases    | EC2/RDS login, WorkSpaces, FSx, IAM Identity Center |

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